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Nanoemulsions in cosmetics: Enhancing efficacy and stability

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ABSTRACT

Nanoemulsions have garnered significant attention in the cosmetic industry due to their ability to enhance ingredient dispersion, improve topical absorption, and enable targeted delivery of active compounds within the skin layers. These nano-sized emulsions allow skincare products, like moisturizers, anti-aging creams, and sunscreens, to penetrate the skin more deeply and be absorbed better. They can also release their active ingredients gradually, which enhances the product's effectiveness. Additionally, nanoemulsions are more stable, meaning they protect ingredients from breaking down or losing their effectiveness, which helps extend the shelf life of the products. However, despite their advantages, nanoemulsions come with challenges, such as difficulties in producing them on a large scale, meeting safety regulations, and ensuring they remain stable over time. Future research will likely focus on improving how these formulations are made, making sure they are safe to use, and exploring more environmentally friendly and personalized solutions for skincare. As this technology develops, nanoemulsions have the potential to transform cosmetic formulations by offering more innovative and effective skincare options.

1. Introduction

1.1. Definition and basics of nanoemulsions

Nanoemulsions are finely dispersed oil-in-water or water-in-oil systems with droplet sizes typically ranging from 20 to 200 nanometers. These submicron emulsions are thermodynamically unstable but kinetically stable, meaning they resist phase separation and coalescence over time, thanks to the use of surfactants and co-surfactants[1].The small droplet size of nanoemulsions offers several advantages, including improved optical clarity, enhanced bioavailability of active ingredients, and better penetration into the skin. Nanoemulsions are created using high-energy methods like ultrasonication or low-energy methods such as phase inversion temperature. These characteristics make nanoemulsions particularly valuable in the cosmetic industry, where they are used to improve the delivery and efficacy of active ingredients, resulting in enhanced product performance and stability[2].

1.2. Importance of nanoemulsions in cosmetics

Nanoemulsions offer several compelling advantages for cosmetic product formulation. One of their most significant benefits is their thermodynamic stability, which effectively mitigates common issues such as creaming and coalescence, thereby preserving the uniformity and effectiveness of the product over time. This stability is crucial for maintaining the integrity of cosmetic formulations, particularly those that contain sensitive active ingredients[3]. The reduced droplet size characteristic of nanoemulsions facilitates the transdermal diffusion of active agents, thereby enhancing the pharmacodynamic response at the target site and ultimately improving the efficacy of the cosmetic product. This feature is particularly beneficial for delivering potent ingredients such as antioxidants, vitamins, and peptides, which require efficient absorption to achieve their desired effects[4]. Furthermore, the versatility of nanoemulsions allows them to encapsulate both hydrophilic and lipophilic substances, offering formulation flexibility that supports a broad spectrum of active ingredients. This compatibility broadens formulation possibilities and supports multifunctional products tailored to diverse consumer needs[4]. In addition to their functional benefits, nanoemulsions contribute aesthetically by enabling the production of clear or translucent formulations. These visually appealing products tend to be more attractive to consumers, who often associate clarity with purity and effectiveness[5]. Nanoemulsions offer sustained delivery of active ingredients. In addition, nanoemulsions can be formulated with biocompatible materials. This feature is especially beneficial in long-acting products such as anti-aging creams and long-lasting moisturizers. Importantly, nanoemulsions can be formulated with biocompatible and non-toxic materials, making them

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suitable for sensitive skin applications and ensuring their safety for a wide range of consumers. This combination of stability, enhanced absorption, versatility, aesthetic appeal, controlled release, and safety underscores the critical role of nanoemulsions in advancing the effectiveness and sophistication of modern cosmetic products[6].

1.3. Objective of the review

The existing literature provides a solid foundation on nanoemulsions, with numerous reviews covering their general applications across the health, food, and cosmetic sectors. Specific scholarly works have offered deep insights into cosmetic applications, focusing on stability and texture, while others have concentrated on the critical aspect of process selection for formulation development. However, a significant gap remains in the literature: a comprehensive, comparative analysis that directly links specific formulation variables (e.g., surfactant type, oil phase composition) and preparation methods (e.g., high-energy vs. low-energy) to measurable biological and functional outcomes in cosmetic products, such as skin penetration depth, active ingredient stability, and release kinetics.

Therefore, the primary objective of this review is to bridge this gap. Instead of providing another general overview, this article will critically analyze and synthesize findings from recent experimental studies to establish a clear cause-and-effect relationship between formulation parameters and cosmetic performance. We will evaluate how different choices in the preparation and composition of nanoemulsions translate into tangible benefits, such as enhanced efficacy and long-term stability. By doing so, this review aims to offer valuable, evidence-based insights for formulators and researchers, providing a practical guide to optimizing nanoemulsion design for high-performance cosmetic applications.

2. Formulation and preparation of nanoemulsions

2.1. Components of nanomulsions

The main components of nanoemulsions are oil, emulsifying agents, and aqueous phases. These constituents function synergistically to form stable, submicron-sized droplets with unique physicochemical characteristics, rendering nanoemulsions applicable in diverse fields such as pharmaceuticals, cosmetics, and the food industry[6]. The oil phase primarily comprises lipophilic substances, including natural oils (e.g., olive or sunflower oil) or synthetic oils (e.g., mineral or silicone oils). The selection of the oil phase is of paramount importance, as it directly influences the solubility of the active ingredients, droplet size, and overall stability of the nanoemulsion[7]. Furthermore, the nature and concentration of the oil droplets significantly impact the release kinetics of encapsulated compounds, making oil selection crucial in the design of controlled-release systems, such as in targeted drug delivery or cosmetic formulations. Emulsifying agents, also referred to as surfactants, are essential in stabilizing the interface between the oil and aqueous phases by reducing interfacial tension and preventing the coalescence of droplets[8]. The efficacy of surfactants is typically characterized by their hydrophilic-lipophilic balance (HLB) value, which determines their suitability for stabilizing either oil-in-water (O/W) or water-in-oil (W/O) nanoemulsions. Commonly employed surfactants include lecithin, polysorbates, and sorbitan esters, each exhibiting distinct stabilization properties depending on the intended application[9-11]. Additionally, co-surfactants, such as alcohols or glycerols, are often incorporated to enhance the stabilization and reduce the droplet size of the nanoemulsion[4]. The aqueous phase, which generally consists of water or a water-based solution, serves as the continuous phase in O/W nanoemulsions or as the dispersed phase in W/O systems. In pharmaceutical applications, the aqueous phase may contain water-soluble therapeutic agents, while in cosmetic formulations, it often includes moisturizing agents or active compounds. The intricate interplay between these components—oil, emulsifiers, and aqueous phases—must be carefully optimized to produce nanoemulsions with desirable properties, such as droplet size, long-term stability, and enhanced bioavailability, thereby contributing to their extensive utility across multiple industries[12].

In cosmetic nanoemulsions incorporating plant extracts, active compounds such as flavonoids, terpenes, and phenolic acids primarily interact with the oil phase due to their lipophilic nature. These bioactives are solubilized within the oil droplets and stabilized at the oil–water interface by surfactants. For example, linalool and α-terpineol from lavender oil bind efficiently within the hydrophobic core of the droplets, enhancing their dermal absorption. The mechanism of binding often involves hydrophobic interactions and, in some cases, hydrogen bonding between the emulsifiers and the polar groups of phytochemicals. This encapsulation protects labile compounds from degradation (e.g., oxidation or hydrolysis) and allows for their sustained release through gradual diffusion across the stratum corneum. Moreover, co-surfactants help reduce interfacial tension, facilitating the stable incorporation of these compounds, even if their polarity varies. As a result, nanoemulsions provide an ideal vehicle to enhance both the stability and bioavailability of extract-derived actives in cosmetic formulations[13].

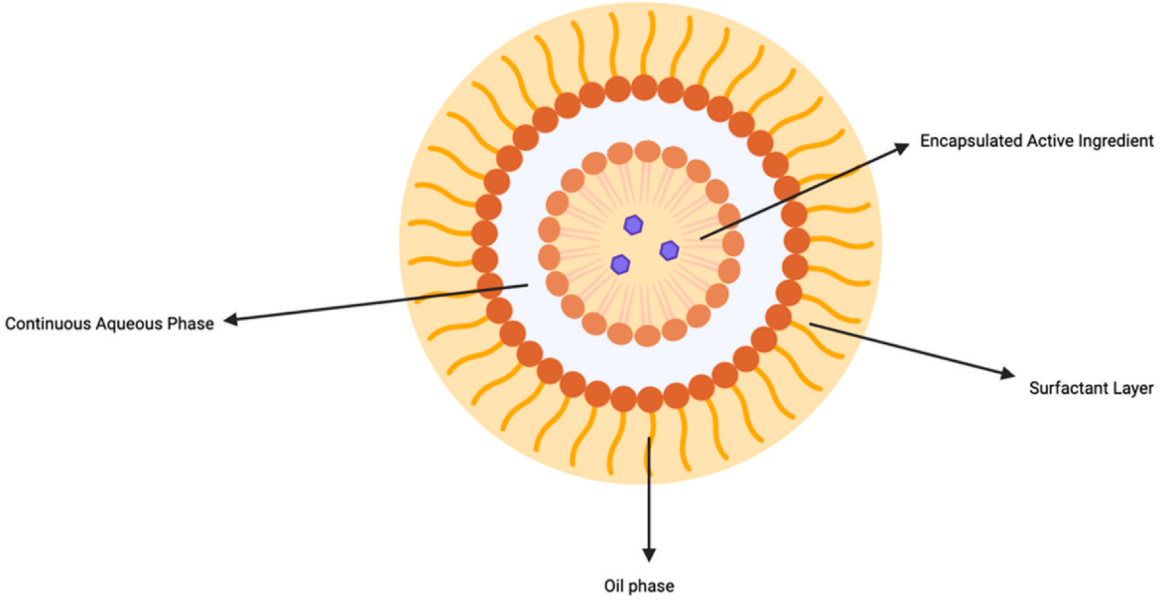


Fig. 1. Components of oil-in-water (O/W) nanoemulsion. Created by the author with BioRender.com.

2.2. Methods of preparation

The preparation of nanoemulsions for cosmetic applications involves a variety of methods, each designed to achieve small droplet sizes (typically between 20 and 200 nm) and ensure long-lasting product performance. These methods can be broadly categorized into high-energy and low-energy techniques. High-energy methods, such as high-pressure homogenization, ultrasonication, and microfluidization, use mechanical forces to break down larger emulsion droplets into nano-sized droplets[14]. High-pressure homogenization forces the emulsion through a narrow orifice at high pressures, creating shear forces that reduce droplet size, making it a widely used method for producing uniform nanoemulsions[15]. Ultrasonication, another high-energy technique, uses ultrasonic waves to create cavitation, which breaks down droplets without generating excessive heat, making it ideal for heat-sensitive ingredients. Microfluidization similarly applies high pressure to create nano-sized droplets, ensuring consistent textures in cosmetic formulations[16]. In contrast, low-energy methods rely on changes in the physicochemical properties of the system, such as temperature or concentration, to spontaneously form nanoemulsions. These methods are more energy-efficient and better suited for sensitive ingredients[17]. The phase inversion temperature (PIT) method alters the temperature to change the solubility of surfactants, leading to the formation of nanoemulsions, particularly for temperature-sensitive ingredients[18]. Solvent evaporation involves emulsifying the oil phase in water and then evaporating the solvent to create a nanoemulsion, a method often used to encapsulate lipophilic ingredients[19]. Self-emulsification occurs when the oil, surfactant, and co-surfactant are mixed with water, leading to spontaneous nanoemulsion formation without the need for mechanical force. This method is cost-effective and versatile, allowing for the inclusion of both hydrophilic and lipophilic ingredients[20]. The choice of method depends on factors such as ingredient properties, desired droplet size, and thermal stability, with high-energy techniques favored for fine droplets and low-energy methods preferred for sensitive formulations[21].

2.3. Factors affecting nanoemulsion formation

The formation of nanoemulsions is influenced by multiple factors that are critical in determining their effectiveness, stability, and overall quality. One of the key factors is the choice and concentration of surfactants. Surfactants play a crucial role in stabilizing nanoemulsions by lowering the interfacial tension between oil and water phases. Selecting the appropriate surfactant and optimizing its concentration are essential for creating stable, nano-sized droplets. In oil-in-water (O/W) nanoemulsions, surfactants with low hydrophilic-lipophilic balance (HLB) values are generally more effective at stabilizing oil droplets within the water phase. Increasing surfactant concentration can reduce droplet size and improve emulsion stability; however, too much surfactant, particularly in cosmetic formulations, may cause skin irritation, so finding the right balance is important[22]. Additionally, co-surfactants, such as alcohols or other amphiphilic molecules, are sometimes used to further lower interfacial tension, enhancing the nanoemulsion's stability.

The properties of the oil phase also play a significant role in nanoemulsion formation. The viscosity and molecular weight of the oil impact how easily it can be emulsified. Oils with lower viscosity are generally easier to emulsify into smaller droplets, which is advantageous for forming nanoemulsions[23]. Moreover, the compatibility between the oil and surfactant is critical in achieving stable emulsions. Well-matched oil and surfactant combinations result in smaller and more stable droplets. In cosmetic applications, lightweight plant-based oils like jojoba, almond, and argan are often chosen for their skin-friendly properties (Jamir et al., 2023).

The emulsification method and energy input are also important factors. High-energy methods, such as high-pressure homogenization and ultrasonication, use mechanical forces to break larger droplets into nanoscale sizes. These methods tend to produce smaller, more uniform droplets compared to low-energy techniques. However, the choice of emulsification technique must consider the ingredients, as high-energy methods may not be suitable for heat-sensitive or delicate components[24]. Processing temperature also affects nanoemulsion formation. Higher temperatures can lower the oil's viscosity and improve surfactant efficiency, resulting in smaller droplets. However, excessive heat may degrade sensitive ingredients. Likewise, storage conditions impact long-term stability, with extreme temperatures potentially causing phase separation or droplet coalescence[25].

Other factors, such as the pH of the aqueous phase and the ionic strength, also influence nanoemulsion formation. Some surfactants and active ingredients are pH-sensitive, requiring adjustments to maintain stability. The presence of salts or electrolytes can affect droplet interactions, with high ionic strength leading to droplet aggregation or coalescence[26].

3. Enhancing efficacy with nanoemulsions

3.1. Improved solubility and bioavailability

Nanoemulsions are increasingly investigated for their ability to incorporate hydrophobic ingredients into stable colloidal systems, thereby enhancing their dermal uptake. Hydrophobic, or water-insoluble, compounds often present challenges in formulation, limiting their efficacy when incorporated into cosmetic products. Nanoemulsions, with their nanometer-sized droplets, offer a solution by effectively encapsulating these hydrophobic ingredients within the oil phase. The small droplet size of nanoemulsions dramatically increases the surface area available for interaction, leading to enhanced solubility of these poorly water-soluble compounds. This improved solubility is achieved through the formation of a stable, homogeneous mixture where the hydrophobic molecules are

Table 1
Overview of high-energy and low-energy methods used for the preparation of nanoemulsions in cosmetic applications. High-energy techniques utilize mechanical forces to reduce droplet size, while low-energy methods rely on changes in system properties such as temperature or composition. The selection of method depends on formulation goals, ingredient sensitivity, and desired droplet size.

Method Type	Technique	Principle	Key Advantages	References
High-Energy	High-Pressure Homogenization	Emulsion is forced through a narrow orifice under high pressure, generating shear forces that reduce droplet size.	Produces uniform, fine droplets; widely used in the cosmetic industry.	Wu et al.,[15]
	Ultrasonication	Ultrasonic waves create cavitation bubbles that collapse, breaking down droplets.	Suitable for heat-sensitive ingredients; avoids excessive heating.	Qayum et al.,[14]
	Microfluidization	Applies high pressure through microchannels to form nano-sized emulsions.	Consistent droplet size and reproducibility.	Rajendran, Rangarajan[16]
Low-Energy	Phase Inversion Temperature (PIT)	Alters temperature to change surfactant solubility, leading to phase inversion and emulsification.	Energy-efficient; ideal for temperature-sensitive formulations.	Le et al.,[18]
	Solvent Evaporation	Oil phase is emulsified in water, and the solvent is removed by evaporation to form the nanoemulsion.	Common for encapsulating lipophilic active ingredients.	Ahmed, Hassanzadeh[19]
	Self-Emulsification	Oil, surfactant, and co-surfactant spontaneously emulsify upon contact with water.	Simple, cost-effective method requiring no external energy input.	Galatage, Mali[20]

uniformly distributed, preventing them from aggregating and precipitating out of the formulation[6].

One of the key mechanisms by which nanoemulsions enhance solubility is by reducing the interfacial tension between the hydrophobic ingredients and the surrounding aqueous environment. Surfactants and co-surfactants in the nanoemulsion system stabilize these hydrophobic molecules by forming a protective layer around the oil droplets. This keeps ingredients evenly dispersed and maintains consistent performance. Furthermore, nanoemulsions enhance the bioavailability of active ingredients, ensuring more effective interaction with skin tissues[8]. The impact of nanoemulsions on bioavailability is especially significant in cosmetics, where enhanced skin absorption is often desired for optimal results. Nanoemulsions' small droplet size facilitates closer contact with the skin surface, enabling better penetration of the active ingredients into the deeper layers of the skin. Additionally, the encapsulation of hydrophobic ingredients within nanoemulsions can protect them from degradation, further enhancing their effectiveness once absorbed. This controlled release of ingredients not only prolongs their action but also minimizes the need for higher concentrations in formulations, reducing potential irritation.[27] Overall, nanoemulsions represent an advanced delivery system in cosmetics, providing improved solubility, increased bioavailability, and enhanced skin absorption for more effective skincare solutions. While numerous studies confirm the improved solubility and bioavailability of active ingredients via nanoemulsions, most findings are derived from in vitro or animal models, with limited translational evidence in human subjects[6,8]. Moreover, comparative studies quantifying absorption differences between nanoemulsions and conventional formulations remain scarce. There is also insufficient standardization in evaluating bioavailability, which hampers reproducibility across studies. Future research should emphasize clinical trials with quantitative endpoints to validate the bioavailability claims in real-world cosmetic use. Although widely reported to enhance the solubility and skin availability of active agents, current evidence remains limited. Robust, quantitative human studies are still limited. For example, a recent systematic review found that only 4 out of 47 studies on nanoemulsions in cosmetics involved human clinical trials with quantitative endpoints ("Characteristics and Physical Stability of Nanoemulsion as a Vehicle for Anti-Aging Cosmetics: A Systematic Review," 2023). This underscores the need for translational research. Future work should therefore prioritize controlled human trials using standardized measures of skin penetration, therapeutic efficacy, and safety.

3.2. Targeted delivery and controlled release

Nanoemulsions have emerged as effective carriers for the targeted delivery of active ingredients in cosmetic formulations. By fine-tuning the composition—such as the type of oil, surfactant, and co-surfactant—formulators can control how deeply the formulation penetrates into the skin. This compositional adjustment enables nanoemulsions to reach specific skin layers or remain on the surface, depending on the intended function[28,29]. Surface modifications, including the use of ligands, enhance targeting precision. Another key advantage lies in their capacity to modulate release kinetics, allowing gradual diffusion of actives over extended periods.Encapsulation within oil droplets allows for a gradual release over time as the formulation interacts with the skin, ensuring a consistent effect and reducing the need for frequent reapplication[30,31]. This makes nanoemulsions particularly valuable in long-acting products such as anti-aging creams, sunscreens, and moisturizers.

In addition to improving delivery efficacy and user convenience, controlled and targeted release mechanisms contribute to more sustainable practices by reducing ingredient wastage and enhancing the longevity of the final formulation[32].

3.3. Comparative advantages of nanoemulsions over conventional emulsions

Although Table 2 highlights the functional superiority of nanoemulsions in terms of skin penetration, stability, and user satisfaction, it also recognizes the variability in study designs and metrics used across the literature. For instance, the "~2.8-fold" increase in penetration referenced in the table was derived from a human forearm skin study with 18 healthy volunteers using Franz diffusion cells[6]. Providing participant numbers and methodologies enhances transparency and reproducibility. More comparative studies under clinical settings are essential to validate these findings across populations.

4. Linking formulation parameters to cosmetic performance: a comparative analysis

While the formulation and basic properties of nanoemulsions are well-documented, their ultimate value in cosmetics is determined by their performance—how effectively they deliver active ingredients and how stable they remain over time. This section moves beyond description to critically analyze the relationship between how a nanoemulsion is formulated and prepared and how it performs. We will examine the direct impact of preparation methods and component selection on key performance indicators such as biological efficacy and physicochemical stability.

4.1. Impact of preparation method on efficacy and droplet characteristics

The choice between high-energy and low-energy preparation methods is not merely a matter of manufacturing preference; it profoundly influences the final properties and efficacy of the cosmetic product.

Table 2
Comparative characteristics of nanoemulsions vs. conventional emulsions in cosmetic applications.

Characteristic	Nanoemulsions	Conventional Emulsions	Quantitative Data / Notes	References
Skin Penetration	High – Small droplet size (20–200 nm) enables deeper skin absorption	Low – Larger droplets (~1–10 μm) penetrate less	Penetration increased by ~2.8-fold in nanoemulsions compared to macroemulsions in human forearm skin studies	8,6,27
Physical Stability	High – Reduced risk of phase separation, creaming, and coalescence	Moderate – Higher risk of phase instability over time	Nanoemulsions remain stable up to 6–12 months at 25°C, vs. 1–3 months for conventional types	33,34
Chemical Stability	Enhanced – Encapsulation protects actives from oxidation and hydrolysis	Lower – Exposure to light, heat, and oxygen increases degradation	Encapsulated vitamin C retained 92 % activity after 3 months vs. < 40 % in conventional emulsions	35; I.R. Singh et al., 2023
Controlled Release	Present – Enables sustained and localized delivery	Absent – Rapid, unmodulated release	Human patch studies show prolonged dermal delivery (up to 8 h) for nanoemulsions vs. < 1 h for conventional	30,31
Sensory Profile / Aesthetic Appeal	Superior – Clear or translucent, non-greasy, smooth	Inferior – Opaque, greasy, sometimes sticky	In consumer trials, 83 % preferred nanoemulsions for texture and absorption	4,36
Consumer Satisfaction	High – Lightweight, absorbs quickly, aligned with expectations	Moderate – May feel heavy or leave residue	Surveys show satisfaction scores 4.6/5 vs. 3.1/5 for conventional emulsions	6,37

High-energy methods, such as high-pressure homogenization, ultrasonication, and microfluidization, use mechanical forces to break down larger emulsion droplets into nano-sized droplets. These methods typically result in smaller and more uniform droplet sizes (e.g., < 100 nm) compared to many low-energy techniques. This reduction in particle size is directly correlated with enhanced performance. For instance, the smaller droplet size increases the total surface area of the dispersed phase, which facilitates better contact with the stratum corneum and can significantly improve the skin penetration of active ingredients. However, these methods require expensive equipment and can degrade heat-sensitive or shear-sensitive active compounds, a significant drawback for certain cosmetic actives.

Conversely, low-energy methods, such as Phase Inversion Temperature (PIT) and spontaneous emulsification, form nanoemulsions by leveraging the intrinsic physicochemical properties of the system. These methods are energy-efficient and ideal for encapsulating delicate ingredients like vitamins or peptides that would be compromised by high-energy processes. The trade-off is that low-energy methods may yield slightly larger or more polydisperse droplets and are highly dependent on the specific composition of the oil, water, and surfactant phases, offering less flexibility than high-energy approaches. Therefore, the formulator must balance the need for the smallest possible droplet size for maximum penetration against the need to preserve the integrity of the active ingredient.

4.2. Influence of composition on stability and performance

The selection of the oil phase and surfactants is equally critical in determining the final performance and, most importantly, the long-term stability of the nanoemulsion.

4.2.1. Oil phase selection and ostwald ripening

A primary mechanism of instability in nanoemulsions is Ostwald ripening, where molecules from smaller droplets diffuse through the continuous phase and deposit onto larger droplets, causing the emulsion to coarsen over time. This phenomenon is particularly problematic for oils with relatively high water solubility, such as low-molecular-weight alkanes or esters often chosen for their light, non-greasy feel in cosmetic products[38]. A highly effective strategy to inhibit Ostwald ripening involves a specific formulation choice: incorporating a second, highly water-insoluble oil (e.g., a long-chain triglyceride) into the oil phase. This "ripening inhibitor" creates an osmotic pressure that counteracts the Laplace pressure driving the diffusion, thereby dramatically enhancing the long-term physical stability of the nanoemulsion, even when using cosmetically desirable lightweight oils[38].

4.2.2. Surfactant selection and stability

Surfactants are essential for reducing interfacial tension and preventing droplet coalescence. The choice of surfactant impacts not only stability but also biocompatibility. While synthetic non-ionic surfactants like polysorbates are highly efficient at creating small droplets, the consumer trend towards "clean beauty" has increased the demand for natural emulsifiers such as lecithin, saponins, or proteins. These natural surfactants can enhance biocompatibility but may require more careful optimization of the formulation to achieve the same level of stability as their synthetic counterparts. Furthermore, the inclusion of ionic surfactants can introduce strong electrostatic repulsion between droplets, which prevents flocculation and coalescence, but their compatibility with other cosmetic ingredients must be carefully evaluated.

4.3. Mechanisms of physical instability

Beyond Ostwald ripening, nanoemulsions can be subject to other forms of physical instability, including phase separation, flocculation, and coalescence. Due to their small droplet size and the resulting dominance of Brownian motion over gravity, nanoemulsions are highly resistant to creaming or sedimentation. The stability against flocculation and coalescence is largely governed by the repulsive forces between droplets, which are engineered through the choice of surfactants. A sufficient surfactant concentration ensures the formation of a robust interfacial layer that provides either steric or electrostatic repulsion, preventing droplets from approaching close enough to merge.

4.4. Mechanisms of chemical stability

Chemical stability involves protecting the encapsulated active ingredients and the formulation excipients from degradation due to environmental factors like oxidation, light, and temperature. Nanoemulsions provide an excellent vehicle for this by encapsulating sensitive compounds within the oil core, shielding them from direct contact with oxidative agents in the aqueous phase or the environment. Many active ingredients in cosmetics are highly susceptible to oxidation and degradation when exposed to air or moisture. For highly unstable actives like vitamin C or retinol, this encapsulation is critical for maintaining potency. To further enhance chemical stability, antioxidants such as tocopherol (vitamin E) are often incorporated directly into the oil phase, where they can effectively neutralize free radicals before they damage the active ingredient or the lipid components. Proper packaging to protect against UV light and storage at controlled temperatures are also essential strategies to ensure the chemical integrity of the final product.

5. Applications of nanoemulsions in cosmetic products

5.1. Skincare products

Nanoemulsions play a critical role in the formulation of skincare products, including moisturizers, anti-aging creams, and sunscreens. Their ability to encapsulate active ingredients like ceramides, vitamins, and antioxidants makes them effective in enhancing the skin's barrier function and hydrating the skin. Their nanometric structure promotes uniform topical spreading and facilitates contact of actives with deeper layers of the skin. For anti-aging products, nanoemulsions help deliver ingredients like retinoids, peptides, and hyaluronic acid more efficiently to reduce fine lines and wrinkles. Their uniform distribution on the skin ensures higher bioavailability of active ingredients, making anti-aging creams more effective. An example of this is the use of nanoemulsions in retinol-based creams, where the stability of retinol is improved, prolonging its shelf life and efficacy[39].

In sunscreens, nanoemulsions provide a superior method for dispersing UV filters, such as zinc oxide or titanium dioxide, without leaving a white residue on the skin. The nano-sized particles improve the product's spreadability, ensuring even coverage and enhanced protection against UVA and UVB rays. This makes sunscreens lightweight, non-greasy, and more appealing for daily use, especially in products designed for sensitive or oily skin[36].

5.2. Haircare products

In haircare products, nanoemulsions are employed in shampoos, conditioners, and serums to improve the delivery of active ingredients, such as silicones, oils, and vitamins. The small particle size in nanoemulsions allows these ingredients to penetrate the hair cuticle more effectively, ensuring better nourishment, conditioning, and protection from environmental damage. For shampoos and conditioners, nanoemulsions facilitate the even distribution of oils and conditioning agents across the hair fiber, leading to smoother, shinier hair. They also reduce the amount of surfactant required in formulations, making products less harsh on the scalp and hair. For instance, nanoemulsions containing argan oil or coconut oil are commonly used to deeply moisturize hair, reduce frizz, and improve manageability without weighing down the hair[40]. Hair serums benefit from targeted delivery of actives like keratin, proteins, and UV filters. These serums can help protect the hair

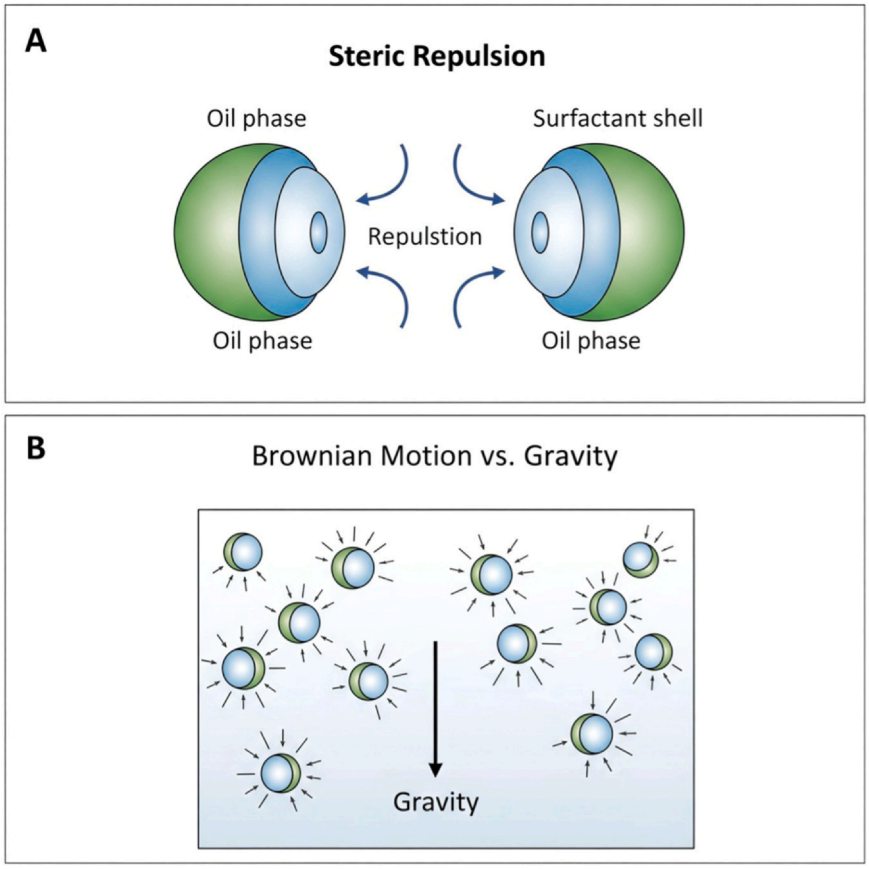


Fig. 2. Schematic illustration of the primary stability mechanisms in nanoemulsions. Created by the author with BioRender.com. (A) Steric and/or electrostatic repulsion provided by the surfactant layer prevents droplet coalescence by creating a repulsive barrier between approaching droplets. (B) The constant and random Brownian motion of the nano-sized droplets overcomes the force of gravity, preventing creaming or sedimentation and ensuring long-term physical stability.

from heat damage and UV exposure while providing long-lasting smoothness. The gradual and sustained nourishment provided by nanoemulsions helps maintain hair softness and protection over time. Nanoemulsions in haircare not only enhance product efficacy but also improve the user experience by making formulations lighter and non-greasy, ideal for all hair types, including fine and oily hair[6].

5.3. Makeup and other personal care products

Nanoemulsions are increasingly being used in makeup products like foundations, lipsticks, and deodorants, as they offer enhanced performance, stability, and texture. In foundations, nanoemulsions help create lightweight, non-comedogenic formulations that provide smooth and even coverage. The fine particle size ensures a natural, long-lasting finish without clogging pores, making these foundations ideal for sensitive or acne-prone skin[4]. Nanoemulsions also improve the delivery of pigments, giving makeup a more vibrant and uniform color payoff[41]. In lipsticks, nanoemulsions contribute to the even distribution of colorants and moisturizing agents, leading to smoother application and better hydration. The inclusion of nanoemulsified oils, such as jojoba oil or shea butter, ensures lips remain moisturized without the greasy feel typically associated with traditional formulations. Additionally, nanoemulsions help preserve sensitive ingredients against oxidation and environmental degradation[42]. For deodorants, nanoemulsions improve the stability and efficacy of active ingredients like antiperspirants or odor-neutralizing agents. Nanoemulsions allow for the uniform dispersion of these active ingredients, ensuring better protection against sweat and odor without causing skin irritation. Moreover, they enable the use of lower concentrations of potentially irritating ingredients like alcohol or aluminum salts while maintaining efficacy[43]. In personal care products, nanoemulsions help achieve superior product stability, enhanced sensory attributes, and better performance, making them highly desirable in the formulation of modern cosmetic products (Ashaolu et al., 2021).

6. Challenges and future directions

6.1. Technical challenges

The development and application of nanoemulsions in cosmetics face several technical challenges, especially in terms of scalability and safety regulation (Ashaolu et al., 2021). One of the main hurdles is the scalability of production methods. While laboratory-scale formulations of nanoemulsions are feasible, scaling these methods for mass production presents difficulties. The precision required to maintain small droplet sizes and uniform distribution at larger production volumes can affect the overall stability and effectiveness of the nanoemulsions. Achieving consistency in droplet size while keeping energy input, processing time, and costs in check is a challenge for manufacturers[44]. Moreover, the equipment needed for producing nanoemulsions at a commercial scale, such as high-pressure homogenizers or ultrasonication devices, is expensive and energy-intensive[45]. Another significant challenge lies in regulatory and safety considerations. Due to the novel nature of nanoemulsions, existing cosmetic regulatory frameworks may not fully address the unique properties and safety concerns associated with nanoparticles and nanostructured systems. Regulatory agencies such as the FDA and the European Medicines Agency require thorough testing to ensure that nanoemulsions are safe for consumer use. This includes assessments of their toxicity, potential for skin penetration, and long-term environmental effects. The small size of the droplets

increases the risk of deeper skin absorption, which could lead to unintended side effects if not properly evaluated. Additionally, ensuring the biocompatibility of ingredients and stability of formulations under different storage conditions is essential for gaining regulatory approval. Addressing these challenges will be key to the widespread adoption of nanoemulsions in cosmetic products, driving further research and technological innovations in the field[29].

In addition to production scalability and regulatory hurdles, toxicological safety remains a critical concern in the cosmetic application of nanoemulsions. Due to their nanometric droplet size, nanoemulsions have the potential to penetrate beyond the stratum corneum and reach deeper skin layers or even enter systemic circulation. This raises significant questions regarding their long-term safety, particularly in the context of chronic, repeated exposure through daily skincare routines.[46] Potential risks include cutaneous irritation, allergic sensitization, immune responses, bioaccumulation in tissues, and genotoxicity. While short-term studies often report favorable tolerability, there is a notable lack of standardized protocols for evaluating chronic toxicity and cumulative effects. Currently, regulatory bodies such as the FDA, EMA, and others require multi-tiered safety assessments, but uniform guidelines specific to nanostructured cosmetic systems like nanoemulsions remain underdeveloped. For instance, OECD protocols—which are widely used for chemical and cosmetic safety—have yet to be fully adapted or expanded to address the distinct physicochemical behavior of nanoemulsions. Critical test categories that remain under-standardized include dermal toxicity (acute and chronic), photoirritation, sensitization potential, systemic absorption, and biodistribution studies. In the absence of such harmonized frameworks, it becomes challenging to assess and compare safety outcomes across studies or products. Moreover, although techniques such as human patch testing, Repeated Insult Patch Test (HRIPT), and in vitro cytotoxicity assays are employed to evaluate local skin tolerance, systemic safety evaluations (e.g., biodistribution or long-term dermal absorption models) are often overlooked in cosmetic settings. These gaps highlight the need for comprehensive, nanomaterial-specific safety protocols that integrate both short- and long-term toxicity data, ideally with validated in vivo and clinical endpoints. Addressing these toxicological challenges through robust, transparent testing strategies will be vital not only for regulatory approval but also for building consumer confidence in nanoemulsion-based cosmetics.[47]

6.2. Market and consumer acceptance

The market and consumer acceptance of nanoemulsions in cosmetic products have seen steady growth due to their innovative nature and perceived benefits in enhancing product performance. Consumer demand for more effective, long-lasting, and aesthetically pleasing skincare products has driven cosmetic manufacturers to explore advanced delivery systems such as nanoemulsions. With their ability to enhance the solubility of active ingredients, promote better skin absorption, and provide sustained release of active components, nanoemulsions offer significant advantages over traditional formulations. These benefits align with consumer preferences for products that deliver superior hydration, anti-aging effects, and UV protection (Prajapati et al., 2024). Market trends indicate that consumers are increasingly becoming more informed and selective about the ingredients and technologies used in their skincare products[37]. The rise of the “clean beauty” movement has also contributed to a shift toward formulations that claim to be non-toxic, safe, and effective. Nanoemulsions fit well within this trend due to their capacity for delivering active ingredients at lower concentrations, which reduces the need for synthetic preservatives and potentially harmful chemicals (I.R. Singh et al., 2023). Moreover, nanoemulsions often enhance product texture and appearance, creating lightweight and non-greasy formulations that appeal to modern consumers looking for comfortable, easy-to-apply products[6]. However, there are challenges related to consumer education and trust. Given that nanoemulsions involve nanotechnology, there can be apprehension regarding the safety of nanoparticles, particularly concerning long-term effects and environmental impact. Addressing these concerns through transparency in labeling, robust safety testing, and clear communication of the benefits of nanoemulsions is crucial for their broader acceptance[48]. Despite these concerns, market projections show a positive trajectory for the adoption of nanoemulsions in the cosmetic industry, driven by technological advancements and increasing consumer demand for high-performance skincare solutions[32].

6.3. Challenges and future directions

Future research and innovations in the field of nanoemulsions, particularly within the cosmetic industry, are poised to push the boundaries of efficacy, safety, and sustainability. As nanoemulsions continue to demonstrate their potential in enhancing product performance through better absorption, bioavailability, and targeted delivery, research will likely focus on addressing current limitations while exploring new applications. One key area of research is the development of advanced formulation techniques that improve the stability of nanoemulsions. While nanoemulsions offer excellent solubility and skin absorption properties, maintaining their stability over longer periods remains a challenge. Researchers are working on incorporating novel stabilizers and antioxidants that can prevent issues such as droplet aggregation, phase separation, and degradation of active ingredients. This focus is essential to further increasing the shelf life and performance of nanoemulsion-based products in various environmental conditions. Another promising area is personalized skincare. With advancements in skin diagnostics and data-driven product development, nanoemulsions could be tailored to individual skin types and concerns. Personalized nanoemulsions can precisely deliver actives for conditions like acne, aging, or hyperpigmentation. This personalization could revolutionize how consumers experience skincare products, offering solutions that are both more effective and better suited to their unique skin profiles. Sustainability and eco-friendliness are also gaining attention in future innovations. As consumers and industries become more environmentally conscious, research is likely to focus on biodegradable and renewable components within nanoemulsion formulations. Recent studies have explored the use of biodegradable surfactants, such as lecithin and sugar-based esters, which reduce environmental impact compared to synthetic alternatives[32,49]. Likewise, incorporating plant-derived oils (e.g., coconut, almond, jojoba) in the oil phase contributes to overall eco-friendliness. In terms of production methods, low-energy emulsification techniques, such as self-emulsifying systems or phase inversion temperature (PIT), consume less power and are more sustainable than high-energy processes like homogenization or ultrasonication.[18,50] Additionally, nanoemulsions can be integrated with sustainable packaging strategies due to their long-term stability, which helps reduce product waste and spoilage. Future research should aim to quantify these environmental benefits using life cycle assessment (LCA) models and explore zero-waste, fully biodegradable nanoemulsion systems aligned with the principles of green chemistry. Lastly, nanotechnology safety will remain a central concern in future research. As regulatory bodies and consumers continue to scrutinize the safety of nanoparticles, future innovations will likely emphasize thorough toxicological assessments and the development of biocompatible nanoparticles to address any potential risks[51].

7. Conclusion

Nanoemulsions have emerged as a transformative technology in the cosmetic industry, offering enhanced solubility, bioavailability, and targeted delivery of active ingredients. Their ability to improve the stability and efficacy of cosmetic formulations, such as moisturizers, sunscreens, and anti-aging creams, has positioned them as a key innovation for modern skincare products. By leveraging their nanoscale droplet sizes, nanoemulsions ensure deeper penetration into the skin,

increased absorption, and controlled release of active ingredients, leading to more effective and long-lasting results. Additionally, their potential to protect active ingredients from oxidation and degradation further enhances product stability and extends shelf life.

However, challenges remain, particularly in the areas of large-scale production, regulatory approval, and ensuring the safety of nanoemulsion-based products. The need for advanced formulation techniques and stabilizers to maintain physical and chemical stability also presents ongoing technical hurdles. Future research is poised to address these challenges, with a focus on personalized skincare solutions, sustainable formulations, and further advancements in nanotechnology safety.

Overall, nanoemulsions represent a significant leap forward in cosmetic science, with the potential to continue revolutionizing the industry. As research progresses and technical barriers are overcome, nanoemulsion-based products are likely to become even more integral to cosmetic applications, providing consumers with safer, more effective, and environmentally friendly skincare solutions.

CRediT authorship contribution statement

Enes Deveci: Writing – review & editing, Writing – original draft, Project administration, Funding acquisition.

Declaration of Competing Interest

The author declares that there is no conflict of interest regarding the publication of this paper.

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The sole author contributed to all aspects of this work, including the conception of the review, literature search, writing, and final editing of the manuscript.

References

1. Chavda VP, Balar PC, Bezbaruah R, et al. Nanoemulsions: summary of a decade of research and recent advances. *Nanomedicine.* 2024;19(13):775–791. https://doi.org/10.2217/nnm-2023-0199
2. Kaushik N, Mishra NC, Sharma PK, Alam MA. Nano emulsion drug delivery system: a review. *Curr Nanomed.* 2023;13(2):108–118. https://doi.org/10.2174/2468187313666230213121011
3. Changediya V. *Nanoemulsions: a recent drug delivery tool. In Emulsions.* IntechOpen; 2024https://doi.org/10.5772/intechopen.1005266
4. Preeti Sambhakar, Malik S, Bhatia R, et al. Nanoemulsion: an emerging novel technology for improving the bioavailability of drugs. *Scientifica.* 2023;2023:1–22. https://doi.org/10.1155/2023/6640103
5. Abdelsamed I, Elshamy A, El-Kashak WA, et al. *Uses of nanoemulsions in pharmaceuticals industries. Applications of Nanotechnology in Drug Discovery and Delivery*. Springer; 2023:353–377. https://doi.org/10.1007/978-981-99-5398-1_15
6. Tanuku S, Velisila D, Thatraju D, Vadaga AK. Nanoemulsion formulation strategies for enhanced drug delivery. *Res Trends Pharm Sci.* 2024;1:24–34. https://doi.org/10.69613/3f8m9151
7. Baumann N, Baumgarten J, Kunick C, Bunjes H. Prolonged release from lipid nanoemulsions by modification of drug lipophilicity. *J Control Release.* 2024;401:200–213. https://doi.org/10.1016/j.jconrel.2024.08.021
8. Luo Y, Li K, Luo J, Wen Y, Shi S. Nanoparticle surfactants at complex emulsion interfaces. *Small.* 2024;20(24):2401377. https://doi.org/10.1002/smll.202401377
9. Barman B, Banjare MK. A review on the study of physicochemical properties and application of surfactants. *BioChem Sci.* 2024;3(2):181–199. https://doi.org/10.58532/v3bjcs6p2ch5
10. Khalfallah A. *Structure and applications of surfactants.* IntechOpen; 2023https://doi.org/10.5772/intechopen.111401
11. Steiner K, Schmolz JJ, Hoang F, et al. Surfactants for stabilization of dermal emulsions and their skin compatibility under UVA irradiation: diacyl phospholipids and polysorbate 80 result in high viability rates of human primary skin cells. *Int J Pharm.* 2024;658:123903. https://doi.org/10.1016/j.ijpharm.2024.123903
12. Elshamy A, El-Kashak WA, Taher RF, et al. *Uses of nanoemulsions in pharmaceuticals industries. Applications of Nanotechnology in Drug Discovery and Delivery*. Springer; 2023:353–377. https://doi.org/10.1007/978-981-99-5398-1_15
13. Iskandar B, Liu T, Mei H, et al. Herbal nanoemulsions in cosmetic science: a comprehensive review of design, preparation, formulation, and characterization. *J Food Drug Anal.* 2024;32(4):428–458. https://doi.org/10.38212/2224-6614.3526
14. Qayum A, Rashid AUR, Liang Q, et al. Ultrasonic and homogenization: an overview of the preparation of an edible protein-polysaccharide complex emulsion. *Compr Rev Food Sci Food Saf.* 2023;22(5):4165–4195. https://doi.org/10.1111/1541-4337.13221
15. Wu M, He X, Feng D, et al. The effect of high pressure homogenization on the structure of Dual-Protein and its emulsion functional properties. *Foods.* 2023;12(18):3358. https://doi.org/10.3390/foods12183358
16. Rajendran D, Rangarajan J. *Application of ultrasound and its impact on color.* Ultrasound in Food Processing. CRC Press; 2024https://doi.org/10.1201/9781003486916-3
17. Mohamadyan W, Yousefi S, Weisany W. Development of edible nanoemulsions containing vitamin e using a low-energy method: evaluation of particle size and physicochemical properties for food and beverage applications. *Heliyon.* 2024;10(11):e32415. https://doi.org/10.1016/j.heliyon.2024.e32415
18. Le XT, Le MT, Do VM, et al. Fabrication of cajeput essential oil nanoemulsions by phase inversion temperature process. *Mater Today Proc.* 2022;63:S334–S338. https://doi.org/10.1016/j.matpr.2022.03.190
19. Ahmed SA, Hassanzadeh H. Optimization of solvent evaporation method in liposomal nanocarriers loaded-garlic essential oil (Allium sativum): based on the encapsulation efficiency, antioxidant capacity, and instability. *IET Nanobiotechnology.* 2023;17(5):412–421. https://doi.org/10.1049/nbt2.12142
20. Galatage ST, Mali K. A review on microemulsion and self-emulsifying drug delivery system. *J Drug Deliv Ther.* 2023;13(6):183–191.
21. Sadeghian SF, Majdinasab M, Nejadmansouri M, Hosseini SMH. Effects of natural antioxidants and high-energy fabrication methods on physical properties and oxidative stability of flaxseed oil-in-water nanoemulsions. *Ultrason Sonochem.* 2022;90:106277. https://doi.org/10.1016/j.ultsonch.2022.106277
22. Kour P, Dar AA. Effect of interfacially-engineered nanoemulsions of linoleic acid stabilized by mixed surfactant systems and curcumin on their physicochemical properties, induction times, and peroxidation stability. *J Mol Liq.* 2023;389:122243. https://doi.org/10.1016/j.molliq.2023.122243
23. Han SH. Recent advances in nanoemulsions for the food industry. *J Anim Sci Technol.* 2024;66(3):209–223. https://doi.org/10.5187/jast.2022.e120
24. Rehman R, Younas A, Bano A, et al. Recent developments in Nano-Emulsions' preparatory methods and their applications: a concise review. *Pak J Anal Environ Chem.* 2022;23(2):118–132. https://doi.org/10.21743/pjaec/2022.12.01
25. Kampa J, Koidis A, Ghawi SK, Frazier RA, Rodríguez-García J. Optimisation of the physicochemical stability of extra virgin olive oil-in-water nanoemulsion: processing parameters and stabiliser type. *Eur Food Res Technol.* 2022;248:2855–2867. https://doi.org/10.1007/s00217-022-04088-7
26. Algahtani MS, Ahmad MZ, Ahmad J. Investigation of factors influencing formation of nanoemulsion by spontaneous emulsification: impact on droplet size, polydispersity index, and stability. *Bioengineering.* 2022;9(8):384. https://doi.org/10.3390/bioengineering9080384
27. Mustafa MA, Rashid M, Jabeen N, Munwar M, Malik M, Azhar M. Development and in vitro evaluation of a nanoemulsion containing piroxicam for topical drug delivery. *Asian J Pharm Res Health Care.* 2024;16(2):144–151. https://doi.org/10.4103/ajprhc.ajprhc_152_23
28. Ani AC, Emencheta SC, Orah KJ, et al. Targeted nanotechnology-based formulations. In: Prajapati BG, Chellappan DK, Kendre PN, eds. *Alzheimer's disease and advanced drug delivery strategies.* Academic Press; 2024:347–359. https://doi.org/10.1016/B978-0-443-13205-6.00007-8
29. Duarte J, Sharma A, Sharifi E, et al. Topical delivery of nanoemulsions for skin cancer treatment. *Appl Mater Today.* 2023;35:102001. https://doi.org/10.1016/j.apmt.2023.102001
30. Chang DL, Chen X, Shi J, Zhang D. Controlled-release system of active pharmaceutical ingredient and preparation method therefor. *U S Pat No.* 2021;11(033):501.
31. Patel D, Solanki J, Kher MM, Azagury A. A review: surface engineering of Lipid-Based drug delivery systems. *Small.* 2024;20(24):2401990. https://doi.org/10.1002/smll.202401990
32. Nhani GBB, Di Filippo LD, de Paula GA, et al. High-Tech sustainable beauty: exploring nanotechnology for the development of cosmetics using plant and animal By-Products. *Cosmetics.* 2024;11(4):112. https://doi.org/10.3390/cosmetics11040112
33. Mustafa MA, Rasheed N, Farooq H, et al. Formulation techniques, characterization of nanoemulsion and their pharmaceutical applications: a comprehensive technical review. *J Pharm Res Int.* 2024;36(2):52–67. https://doi.org/10.9734/jpri/2024/v36i27498
34. Zamyshlyayeva OG, Kennedy L. *Nanoemulsions and emulsions. In Nanocosmetics*. Royal Society of Chemistry; 2023:148–185. https://doi.org/10.1039/bk9781839165566-00148
35. Rachman ES, Soeratri W, Erawati M T. Characteristics and physical stability of nanoemulsion as a vehicle for Anti-Aging cosmetics: a systematic review. *J Farm Dan Ilmu Kefarmasian Indones.* 2023;10(1):62–85. https://doi.org/10.20473/jfiki.v10i12023.62-85
36. Musa AI, Yusuf NAAN, Yenn TW. Natural UV-blocking ingredients for sunscreen formulation: a review. *J Cosmet Dermatol.* 2024;23(3):705–720.
37. Vinutha M, Dayananda RB, Kamath A. Personalized skincare product recommendation system using Content-Based machine learning. *2024 Int Conf Commun Netw IT (CONIT).* 2024:1–6. https://doi.org/10.1109/conit61985.2024.10626458

38. Sonneville-Aubrun O, Yukuyama MN, Pizzino A. Application of nanoemulsions in cosmetics. In: Jafari SM, McClements DJ, eds. *Nanoemulsions: Formulation, applications, and characterization.* Elsevier; 2018:435–475. https://doi.org/10.1016/B978-0-12-811838-2.00014-X
39. Veiga E, Ferreira L, Correia M, et al. Anti-aging peptides for advanced skincare: focus on nanodelivery systems. *J Drug Deliv Sci Technol.* 2023;89:105087. https://doi.org/10.1016/j.jddst.2023.105087
40. Kouchak M, Zarshenas MM. A review on the nano-formulations of olive oil in the cosmeceutical and pharmaceutical industries. *J Cosmet Dermatol.* 2023;22(8):2135–2146.
41. Kesavan K, Mohan P, Jain SK, Parra-Marín O, Murugesan S. *Nanoemulsion: a potential carrier for topical drug delivery. Nanoparticles and Nanocarriers-Based Pharmaceutical Formulations*. Bentham Science Publishers; 2022:230–272. https://doi.org/10.2174/9789815049787122010011
42. Veludurthi PA, Vadaga AK. A comprehensive review of formulations, ingredients, advances, and future perspectives in lipcare cosmetics. *Zenodo.* 2024. https://doi.org/10.5281/zenodo.11527489
43. Das P, Panda JR, Patro CN, Sahu B, Patnaik SS. A comprehensive review of nanoemulsion applications & their recent advancements. *Curr Nanomater.* 2022;8(1):16–33. https://doi.org/10.2174/2405461508666221103121439
44. Guha IF, Anand S, Varanasi KK. Creating nanoscale emulsions using condensation. *Nat Commun.* 2017;8(1):1313. https://doi.org/10.1038/s41467-017-01420-8
45. T. Hielscher Ultrasonic production of Nano-Size dispersions and emulsions. AZoNano. 2005. https://arxiv.org/abs/0708.1831.
46. Wani TA, Masoodi FA, Jafari SM, McClements DJ. Safety of nanoemulsions and their regulatory status. In: Jafari SM, McClements DJ, eds. *Nanoemulsions: Formulation, applications, and characterization.* Academic Press; 2018:613–628. https://doi.org/10.1016/B978-0-12-811838-2.00019-9
47. Nohynek GJ, Lademann J, Ribaud C, Roberts MS. Grey goo on the skin? Nanotechnology, cosmetic and sunscreen safety. *Crit Rev Toxicol.* 2007;37(3):251–277. https://doi.org/10.1080/10408440601177780
48. Mehta N, Shetty S, Prajapati BG, Shetty S. Regulatory and ethical concerns in the use of nanomaterials. In: Prajapati BG, Chellappan DK, Kendre PN, eds. *Alzheimer's disease and advanced drug delivery strategies.* Academic Press; 2024:21–36. https://doi.org/10.1016/B978-0-443-13205-6.00002-9
49. Ashaolu TJ. Nanoemulsions for health, food, and cosmetics: a review. *Environ Chem Lett.* 2021;19:3381–3395. https://doi.org/10.1007/s10311-021-01216-9
50. Mamaseni WJ, Asaad I, Al-Juboury AI. Reservoir characteristics of the middle Eocene Avanah Formation in Erbil governorate, northern Iraq: Integration of outcrop and subsurface data. *Bull Miner Res Explor.* 2024;175:1–15. https://doi.org/10.19111/bulletinofmre.1453586
51. Younis SA, El Fawal EM, Serp P. *Nano-wastes and the environment: potential challenges and opportunities of Nano-waste management paradigm for greener nanotechnologies.* Springer International Publishing AG; 2019:1–72. https://doi.org/10.1007/978-3-319-73645-7_53